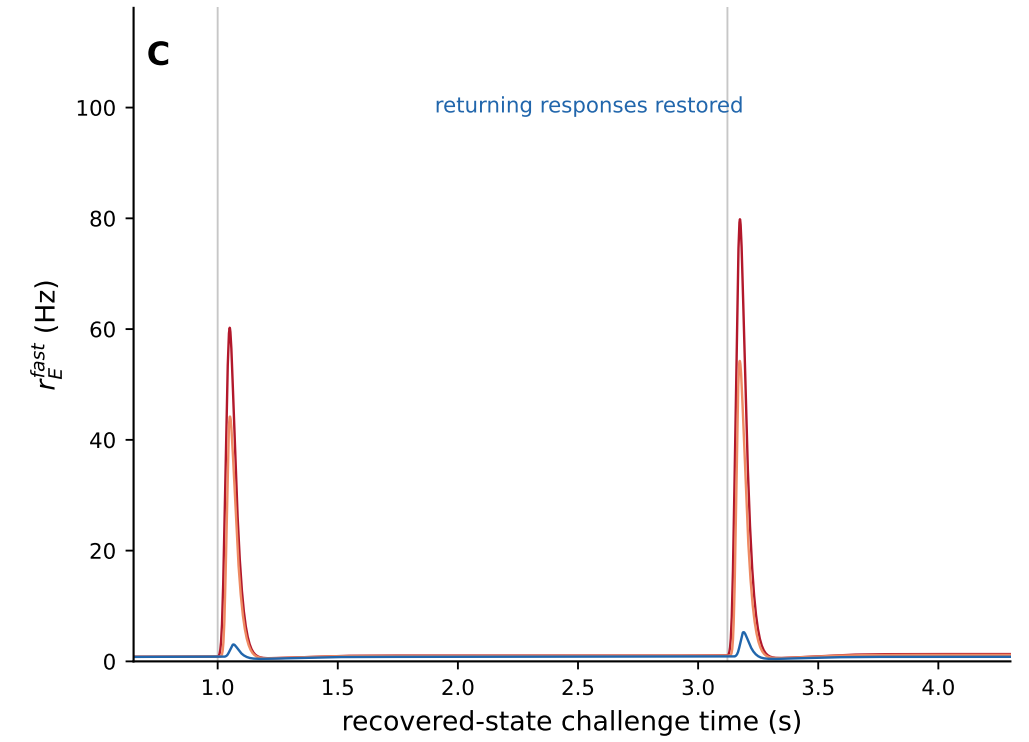
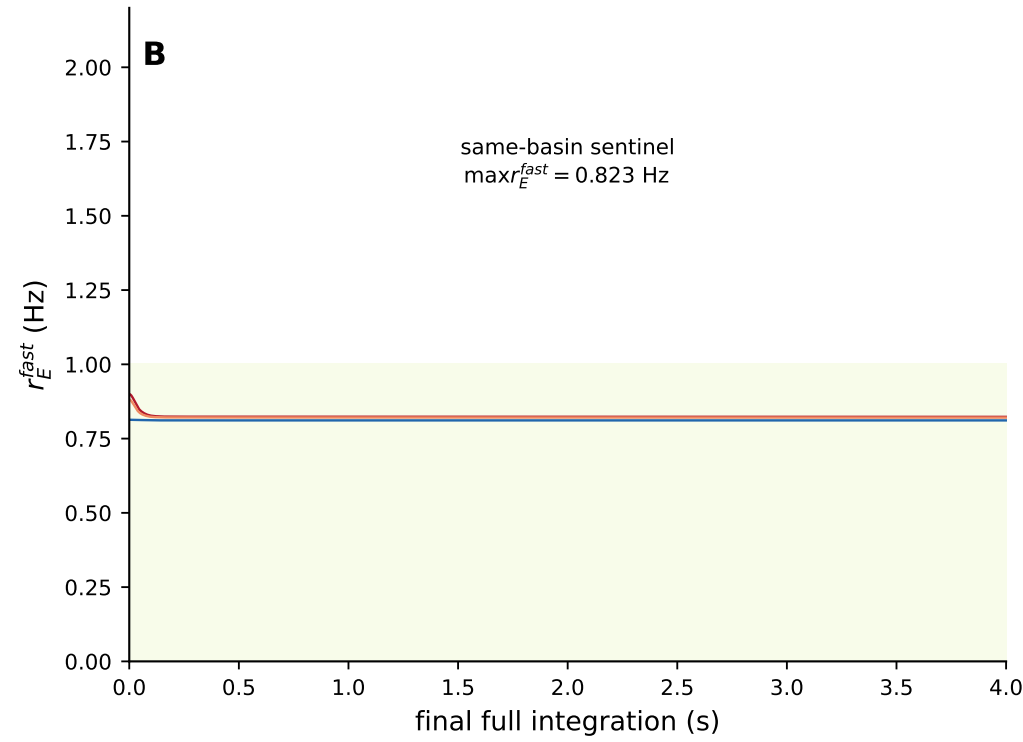
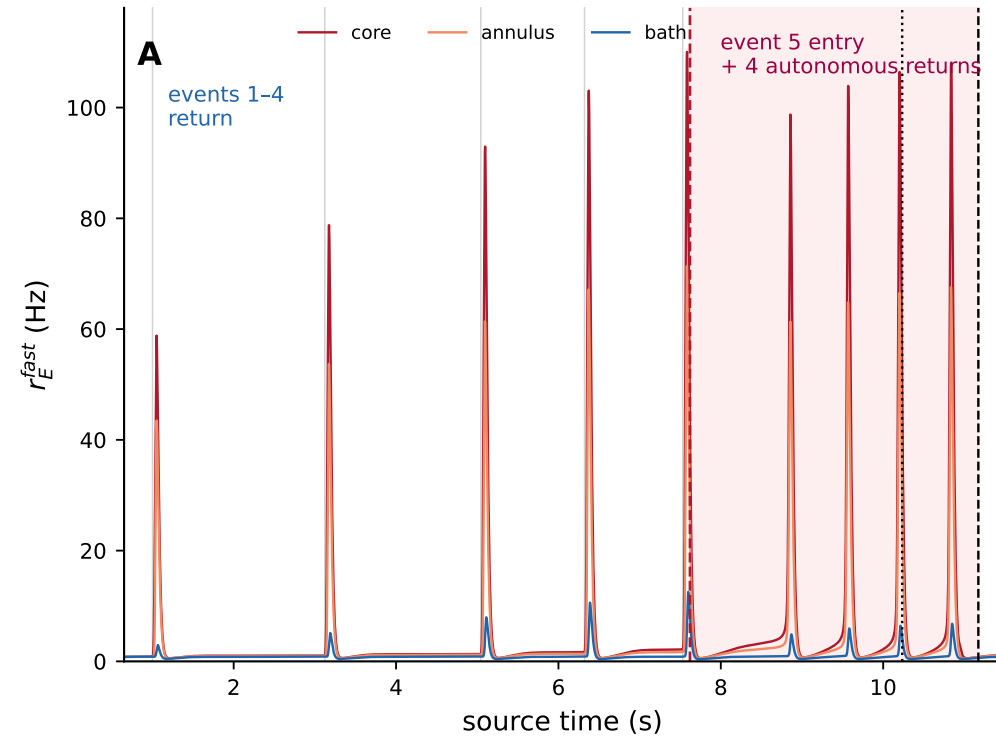


R4 regional q-p-M hybrid lifecycle: fast bursting inside a slow recovery loop

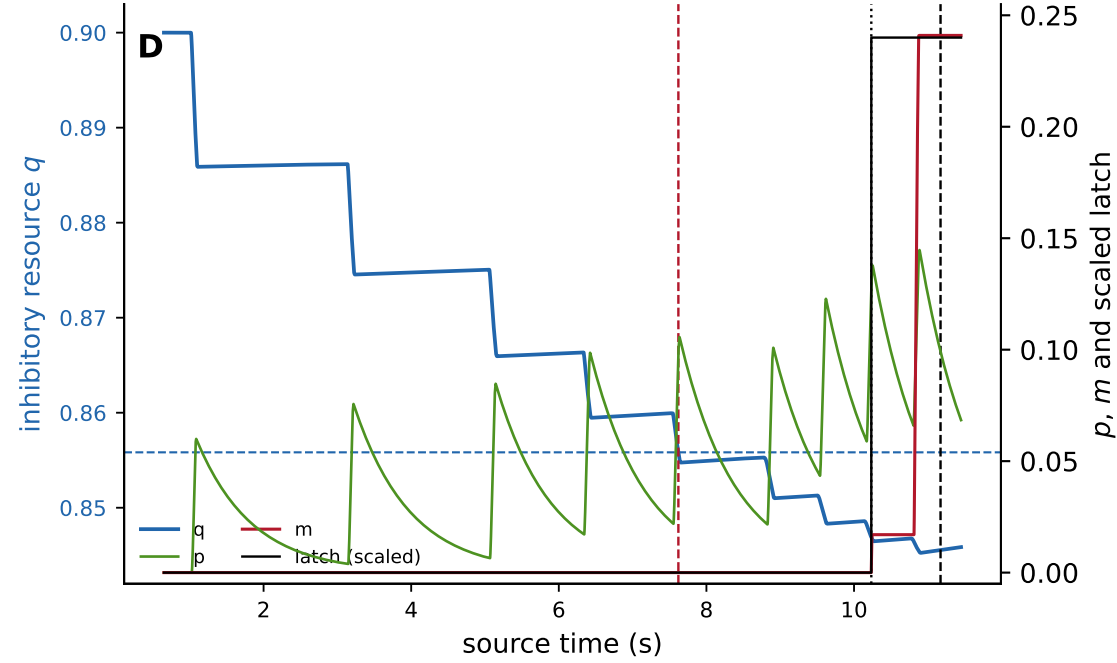
Registered IED-like train → bounded ictal-like bursting → exit

After hybrid recovery: stable low state

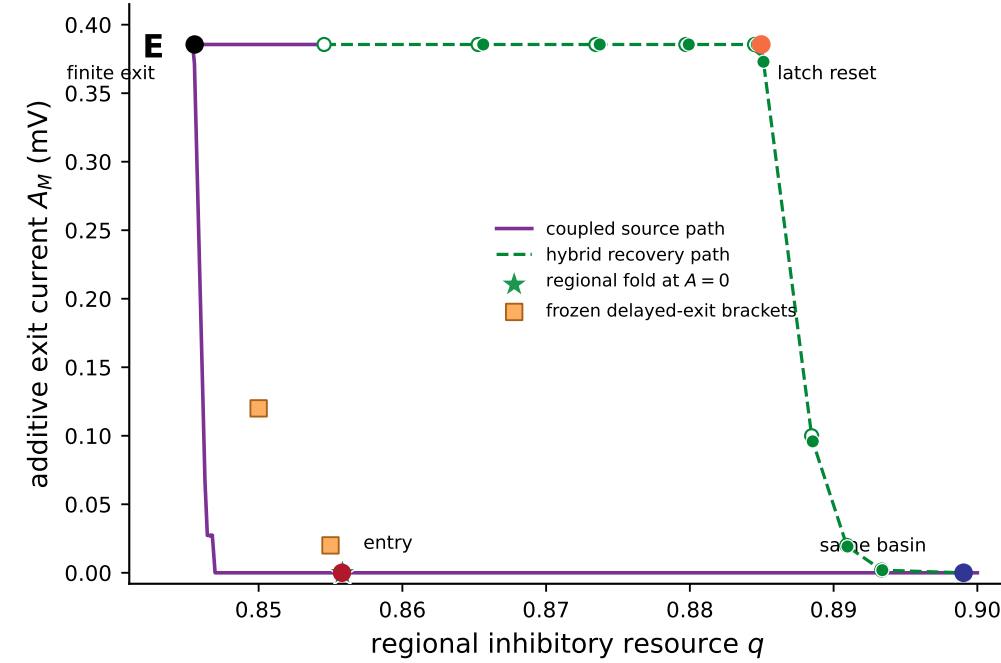
Same-basin fork: interictal-like events resume



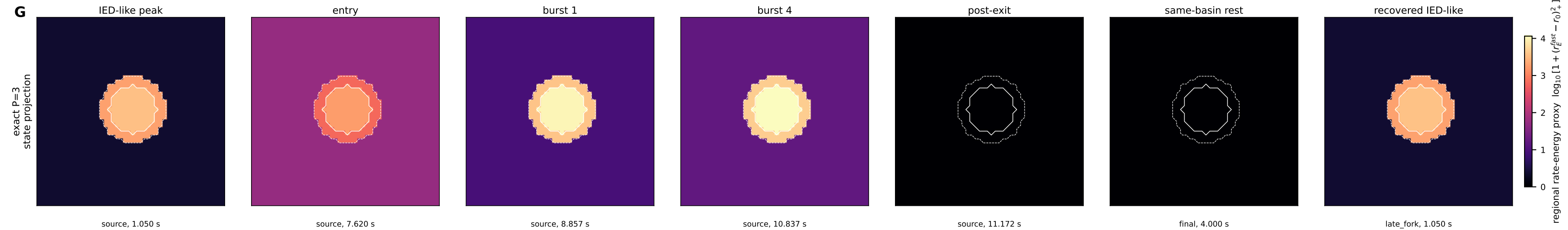
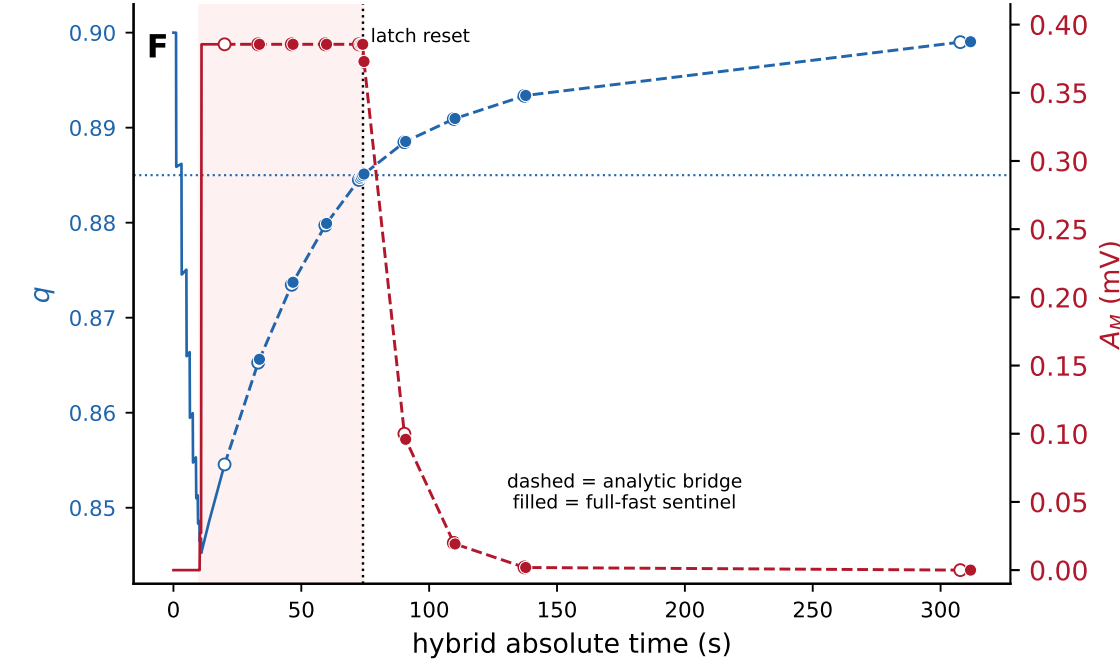
Slow ordering: q opens entry; p sets latch; m supplies exit



Frozen landmarks + actual q-A slow loop



Protected recovery → reset → natural M release



Diagnostic boundary: $P=3$ rate-patch model; registered core pulses trigger entry. Not Virtual-SEEG/LFP, not a continuous field, and the 20–312 s recovery contains analytic bridges plus full-fast sentinels; the late train is a same-basin fork.